Business Case Template

Business Case

DOST STARBOOKS: Whiz Challenge

DOST Science and Technology Information Institute
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APRIL 2026

TABLE OF CONTENTS

1. Executive Summary

The STARBOOKS Whiz Challenge is an enhanced gamified educational platform developed to support the mission of the Department of Science and Technology – Science and Technology Information Institute (DOST-STII) in promoting science, technology, engineering, and mathematics (STEM) education across the Philippines. Designed as a complementary application to the STARBOOKS digital library system, the Whiz Challenge aims to make STEM learning more interactive, engaging, and accessible—particularly for students participating in DOST-led exhibits, outreach programs, and events, including those in areas with limited internet connectivity.

While the original version of the STARBOOKS Whiz Challenge successfully introduced quiz-based gamified learning, it presented limitations in terms of replayability, learning diversity, analytics capability, and overall user experience. To address these gaps, this project focuses on redesigning and enhancing the system by introducing multiple game modes, improved visual design, expanded reward mechanics, performance tracking, and administrative monitoring tools.

The enhanced system now includes four game modes—Whiz Challenge, Whiz Memory Match, Whiz Puzzle, and Whiz Battle—along with a structured badge and star reward system, player leaderboard, and difficulty-based progression. On the administrative side, the system provides account management, quiz configuration, reward monitoring, leaderboard tracking, and analytics dashboards for event-based insights. The application remains offline-first to ensure accessibility in underserved communities, aligning with national education goals and the United Nations Sustainable Development Goals, particularly SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure).

At the time of writing, the project is in its final development phase. All player-side features have been completed, and the majority of the administrative functionalities are operational, with only final analytics refinements remaining. System documentation, including the Software Requirements Specification (SRS), is nearing completion. The next steps involve system polishing, comprehensive testing, user manual preparation, packaging, and deployment.

Through this enhanced version of the STARBOOKS Whiz Challenge, the project aims to provide a scalable, engaging, and data-informed educational tool that strengthens DOST-STII's advocacy for accessible and innovative STEM learning across the country.

1.1. Issue

Despite its success as an introductory gamified platform, the current STARBOOKS Whiz Challenge system presents several operational and educational limitations:

- **Limited Engagement and Replayability:** The system only supports a single quiz mode, leading to repetitive gameplay and declining user interest over time.
- **Lack of Learning Diversity:** The platform does not accommodate different learning styles such as visual, experiential, or competitive learning.
- **Absence of Analytics and Monitoring Tools:** DOST-STII lacks access to player data, engagement trends, and performance insights, limiting evaluation and decision-making.
- **Outdated User Experience:** The system lacks modern design elements, animations, and intuitive navigation expected in contemporary educational applications.

These challenges reduce the platform's effectiveness in achieving its goal of promoting STEM education and engagement.

1.2. Anticipated Outcomes

The enhanced system will:

- **Improved Student Engagement:** Multiple game modes and reward systems will increase replayability and sustained interaction.
- **Enhanced Learning Experience:** Diverse gameplay formats will cater to different cognitive and learning styles.
- **Data-Driven Decision Making:** Built-in analytics will allow DOST-STII to monitor user behavior and optimize outreach strategies.
- **Increased Accessibility:** Offline-first functionality ensures continued usability in remote and underserved areas.
- **Scalability and Sustainability:** The system can be deployed across multiple kiosks and expanded with future enhancements.

The end state is a scalable, engaging, and data-informed educational platform integrated into STARBOOKS kiosks nationwide.

1.3. Recommendation

It is recommended to proceed with the full implementation and deployment of the enhanced STARBOOKS Whiz Challenge system. The proposed solution includes:

- Four interactive game modes (Quiz, Memory, Puzzle, Battle)

- Gamified reward system (badges, stars, leaderboard)
- Player tracking and performance monitoring
- Administrative dashboard with analytics and reporting
- Offline-first architecture with selective online capabilities

This approach addresses all identified limitations while aligning with DOST-STII's mission and operational environment.

1.4. Justification

The project is justified as it:

- Supports national and organizational goals for STEM education and digital inclusion
- Utilizes existing infrastructure and open-source technologies
- Improves educational impact and outreach effectiveness
- Supports SDG 4 (Quality Education) and SDG 9 (Innovation)

Failure to implement the project would result in continued low engagement, limited scalability, and lack of measurable impact.

2. Business Case Analysis Team

The following individuals comprise the business case analysis team. They are responsible for the analysis and creation of the DOST STARBOOKS: Whiz Challenge business case.

Role	Description	Name/Title
Project Sponsor	Provides project oversight and approval	Marievic V. Narquita, ISS Head (DOST-STII)
Project Manager	Leads project execution and coordination	Kelly Dumbrique
Documentation Lead	Handles documentation and reports	Janice Maxene Salipande
UI/UX Designers	Design system interface and experience	Shandrae Lois Quianzon, Arcielle Marie Gercan, Janice Maxene Salipande
Developers	Develop system features and backend	Shandrae Lois Quianzon, Arcielle Marie Gercan
QA/Testers	Conduct system testing and validation	Kelly Dumbrique

3. Problem Definition

3.1. Problem Statement

DOST-STII has long advocated for making science and technology education more accessible and engaging, especially for learners in disadvantaged and underserved areas. Through the STARBOOKS initiative and its outreach activities, one of their key efforts to spark interest among students has been the STARBOOKS Whiz Challenge—a gamified quiz application deployed during science fairs, exhibits, and educational events.

While the current version of the Whiz Challenge has successfully introduced fun, interactive STEM learning through a solo quiz game, DOST-STII has observed several limitations that affect the system's ability to fully meet its engagement and outreach goals:

1. Limited replayability and player retention

As the app currently supports only a single game mode (solo quiz), DOST-STII has noted reduced student engagement over time. Once users have earned badges and rewards, the gameplay becomes repetitive, limiting opportunities for deeper learning or continued use.

2. Lack of diverse learning strategies

The existing format does not accommodate different types of learners. Students who learn better through collaboration, competition, or visual-based games have fewer options to interact with the content in a way that suits their learning style.

3. Difficulty in measuring impact and player insights

Without analytics tools, it is difficult for DOST-STII to gather data about player demographics and behavior, or trends during events. This hinders their ability to refine future outreach strategies or evaluate the educational impact of the app.

4. Need for richer, more modern user experience

As digital learning tools continue to evolve, DOST-STII aims to keep the Whiz Challenge visually appealing and competitive with modern educational apps. However, the current system lacks dynamic visuals, animations, and polish that would better capture the attention of today's learners.

3.2. Organizational Impact

The implementation of the enhanced STARBOOKS Whiz Challenge system will introduce several changes across organizational processes, tools, roles, and technological infrastructure within DOST-STII operations. The following outlines the expected impact:

Tools:

The current quiz-only system will be upgraded into a multi-functional gamified learning platform. This includes the introduction of:

- Multiple game modes (Quiz, Memory, Puzzle, Battle)
- Player leaderboard and statistics tracking
- Admin dashboard with analytics and reporting features

These tools will allow administrators to better manage content, monitor engagement, and evaluate system performance during events and deployments.

Processes:

The project will improve and streamline several operational processes:

- Learning Delivery: Transition from static quiz interaction to dynamic, multi-mode gameplay
- Engagement Monitoring: Introduction of analytics for tracking player behavior and trends
- Content Management: More efficient quiz and reward management through enhanced admin tools
- Event Operations: Better facilitation of activities during exhibits through real-time insights and leaderboard tracking

These improvements will enable more interactive, data-driven, and efficient educational outreach.

Roles and Responsibilities:

The enhanced system will refine and expand user roles:

- Players (Students/Visitors): Gain access to multiple learning modes, performance tracking, and rewards, promoting self-paced and competitive learning.
- Administrators (DOST Staff): Will take on additional responsibilities such as monitoring analytics, managing leaderboards, and overseeing player engagement during events.
- Development Team: Responsible for system maintenance, updates, and future enhancements.

Overall, the system empowers both users and administrators while improving operational efficiency.

Hardware/Software:

The project is designed to remain compatible with the existing STARBOOKS kiosk infrastructure while introducing software enhancements:

- **Hardware:**
  - No additional hardware investment is required
  - Compatible with existing touchscreen all-in-one PCs and standard desktop kiosk setups
  - Supports deployment in schools, libraries, and exhibit venues
- **Software:**
  - Transition to a modern technology stack (Flutter, Laravel, MongoDB)
  - Implementation of Docker for consistent deployment
  - Offline-first architecture with optional online capabilities for multiplayer features

These improvements ensure scalability, maintainability, and improved system performance without increasing operational costs.

3.3. Technology Migration

The STARBOOKS Whiz Challenge enhancement project follows an upgrade and integration approach rather than a complete system replacement. The goal is to improve the existing system by introducing new technologies and features while maintaining compatibility with the current STARBOOKS kiosk infrastructure and ensuring uninterrupted usability in offline environments.

To minimize disruption to ongoing DOST-STII operations and ensure a smooth transition, a phased implementation strategy will be adopted. This approach allows for gradual deployment, testing, and validation of the enhanced system.

Phase I: The enhanced system will be developed using a modern technology stack, including Flutter for the frontend, Laravel for backend services, and MongoDB for data storage. Docker will be utilized to containerize the application components, ensuring consistency across development, testing, and deployment environments.

During this phase, all core functionalities—including multiple game modes, reward systems, player tracking, and admin dashboard features—will be implemented and integrated within a controlled development environment.

Phase II: Once development is completed, the system will undergo rigorous internal testing. This includes:

- Unit testing for individual components
- Integration testing for system modules
- Offline functionality validation
- Performance and usability testing

This phase ensures that all features operate correctly within the offline-first architecture and are optimized for kiosk hardware.

Phase III: Existing quiz content, user data structures, and reward configurations from the current system will be reviewed and prepared for integration into the enhanced platform. System configurations such as difficulty settings, timers, and reward logic will be aligned with the new features. This phase ensures that the enhanced system maintains continuity with existing educational content while enabling new functionalities.

Phase IV: The enhanced system will be deployed in selected STARBOOKS kiosks during controlled environments such as DOST events, exhibits, or pilot schools.

During this phase:

- Real users (students and facilitators) will interact with the system
- Feedback on usability, engagement, and performance will be collected
- Minor adjustments and refinements will be implemented

This step ensures that the system meets user expectations and operates effectively in real-world conditions.

Phase V: Following successful pilot testing, the system will be deployed across STARBOOKS kiosks for official use. The deployment process will include:

- Packaging and installation of the system on kiosk devices
- Final system validation and quality assurance checks
- Distribution of user manuals and administrative guidelines

The legacy version of the Whiz Challenge will be replaced or archived, while ensuring that the enhanced system is fully operational.

Phase VI: After deployment, ongoing support and maintenance will be conducted to ensure system stability and performance. This includes:

- Monitoring system usage and performance
- Addressing bugs and technical issues
- Implementing future enhancements based on feedback and analytics

4. Project Overview

The project involves redesigning the STARBOOKS Whiz Challenge into a multi-functional gamified learning platform. It introduces interactive gameplay, performance tracking, and administrative tools while maintaining compatibility with existing kiosk systems.

4.1. Project Description

The STARBOOKS Whiz Challenge Enhancement Project is an initiative aimed at redesigning and upgrading the existing STARBOOKS Whiz Challenge system developed by DOST-STII. The project focuses on transforming the current single-mode quiz application into a multi-mode, gamified educational platform that enhances student engagement, supports diverse learning styles, and provides data-driven insights for administrators.

The enhanced system introduces four interactive game modes—Whiz Challenge, Whiz Memory Match, Whiz Puzzle, and Whiz Battle—along with an improved reward system, leaderboard functionality, and analytics dashboard. It is designed as an offline-first kiosk-based application, ensuring accessibility for students in remote and underserved areas with limited or no internet connectivity.

Technically, the system utilizes a modern technology stack composed of Flutter (frontend), Laravel (backend), and MongoDB (database), deployed using Docker containers within a local area network (LAN) environment. The solution is compatible with existing STARBOOKS kiosk hardware, enabling seamless integration without requiring major infrastructure changes.

The project supports DOST-STII's mission of promoting STEM education and aligns with national and global initiatives such as SDG 4 (Quality Education) and SDG 9 (Industry, Innovation, and Infrastructure).

4.2. Goals and Objectives

To develop an enhanced, scalable, and engaging gamified learning platform that improves STEM education delivery through the STARBOOKS ecosystem. The following table lists the business goals and objectives that the DOST STARBOOKS: Whiz Challenge Project supports and how it supports them:

Business Goal/Objective	Description
Improve Engagement and Retention	Introduce multiple game modes to reduce repetition and increase replayability Encourage continuous participation through gamification elements
Support Diverse Learning Styles	Incorporate quiz-based, visual, and interactive learning formats Enable both individual and competitive learning experiences
Enable Data-Driven Insights	Provide analytics dashboards for monitoring player behavior and trends Support decision-making for future DOST-STII outreach programs
Enhance User Experience	Redesign the interface with modern UI/UX principles Improve usability for students across different age groups
Maintain Accessibility	Ensure full offline functionality for both players and administrators Support deployment in low-connectivity environments

4.3. Project Performance

This section outlines the key measures that will be used to evaluate the performance and success of the STARBOOKS Whiz Challenge Enhancement Project. These measures focus on how effectively the system improves engagement, streamlines administrative processes, enhances learning delivery, and supports data-driven decision-making.

The performance indicators are aligned with the system's core components, including gameplay experience, administrative efficiency, analytics capability, and system reliability. These will be further quantified and validated during system testing and post-deployment evaluation.

The table below presents the key resources, processes, or services along with their corresponding performance measures and expected outcomes:

Key Resource/Process/Service	Performance Measure
Player Engagement (Game Modes)	Increase in average gameplay sessions per user due to the introduction of multiple game modes (Whiz Challenge, Memory Match, Puzzle, Battle), resulting in improved replayability and sustained user interest.
Learning Effectiveness	Improvement in player performance (e.g., higher quiz scores, faster completion times, increased badge acquisition), indicating better knowledge retention and understanding of STEM topics.
Reward System (Badges & Stars)	Increase in reward claims and badge completion rates, demonstrating higher motivation and continued participation among students.
Leaderboard and Competitive Features	Increased participation in leaderboard rankings and Whiz Battle mode, reflecting enhanced competitive engagement and peer interaction.
Admin Data Management	Reduction in manual tracking and monitoring efforts through automated dashboards, eliminating the need for manual consolidation of player data during events.
Analytics and Reporting	Availability of real-time event insights (e.g., player demographics, most played game mode, reward distribution), reducing reliance on manual reporting and enabling faster decision-making after each event.
System Accessibility (Offline Functionality)	100% availability of core features (games, leaderboard, admin tools) without internet connectivity, ensuring uninterrupted usage in remote or low-connectivity areas.
System Performance and Reliability	Stable system operation during events with minimal downtime or crashes, ensuring smooth gameplay and admin operations across multiple users in a LAN setup.
Administrative Efficiency	Decrease in time required for admin tasks (player management, quiz configuration, reward tracking) due to centralized and user-friendly admin dashboard.
Deployment and Maintenance	Reduced setup and deployment time through Docker-based containerization, ensuring consistent environments and easier system updates across kiosks.
User Experience (UI/UX)	Improved usability and satisfaction based on user feedback, measured through ease of navigation, visual appeal, and interaction responsiveness.

4.4. Project Assumptions

The following assumptions were made during project planning and development:

- Existing STARBOOKS kiosk hardware will remain available and functional
- Users (students and visitors) will have basic familiarity with touchscreen interfaces
- DOST-STII will provide and maintain quiz content and educational materials
- The system will primarily be used in controlled environments (e.g., exhibits, schools, libraries)
- Local area network (LAN) setup will be available for multi-client communication
- Minimal internet connectivity may be available for optional features (e.g., battle mode)
- Admin users will have basic technical knowledge to operate the dashboard

4.5. Project Constraints

The following constraints currently apply to the project:

- Limited IT Resources – The project must share IT staff, hardware, and network resources with other ongoing school technology initiatives, limiting the availability of technical support during development and deployment.
- Software Availability – There are a limited number of commercial off-the-shelf (COTS) products suitable for gamified learning, leaderboard management, and reward tracking, requiring custom development for certain features.
- Internal Implementation – All development, configuration, and deployment will be conducted internally by the project team, without direct support from software vendors or third-party developers, which may affect troubleshooting and system optimization.
- Hardware Constraints – The deployment will rely on existing computer labs and kiosks with fixed specifications, limiting flexibility in system performance and potential upgrades.
- Connectivity Limitations – The system must function in offline mode for areas with limited internet access, which may restrict certain real-time features or cloud-dependent analytics.
- Budget Limitations – Funding is restricted to the resources currently allocated, which constrains the ability to purchase additional software licenses, hardware, or external support.

- Time Constraints – The project must be completed in alignment with the academic calendar, limiting the window for testing, training, and deployment before school events or competitions.

4.6. Major Project Milestones

This section outlines the major milestones for the STARBOOKS Whiz Challenge Enhancement Project along with their preliminary target completion dates. These dates are general estimates for planning purposes and will be refined as the detailed project schedule is developed. Milestones may be adjusted to reflect changes in project scope, resources, or unforeseen challenges.

Milestones/Deliverables	Target Date
Project Charter	April 2025
Project Plan Review and Completion	April – May 2025
Project Kickoff	May – June 2025
Phase I — core player features (login, profile, quiz, stats)	September 2025
Phase II — memory match, puzzle games, and rewards system	October 2025
Phase III — Whiz Battle multiplayer and real-time features	November 2025
Phase IV — admin dashboard (user/content management, AI, analytics)	December 2025
Phase V — system testing, usability/pilot testing, performance optimization	January – March 2026
Code freeze and final deployment to STARBOOKS kiosks	March – April 2026
Soft launch — select schools and DOST events, final QA and dry run	May 2026
Closeout / official release and project completion	June 2026

5. Strategic Alignment

The DOST STARBOOKS: Whiz Challenge supports multiple strategic initiatives by improving student engagement, promoting learning through gamification, and providing actionable analytics for administrators.

Plan	Goals/Objectives	Relationship to Project
2026 Student Learning & Engagement Plan	Enhance student learning and participation	The Whiz Challenge motivates students to actively engage with learning materials through gamified quizzes, rewards, and competitions, reinforcing knowledge retention.

Plan	Goals/Objectives	Relationship to Project
2026 Technology-Driven Education Plan	Utilize digital platforms to improve learning outcomes	The project leverages a web-based quiz platform to deliver real-time challenges, track progress, and provide instant feedback, improving access and responsiveness.
2026 Performance Analytics & Reporting Plan	Improve insights into student performance and program effectiveness	Administrators can monitor participation, track performance trends, and measure engagement metrics, enabling data-driven decisions to improve educational programs.

6. Cost Benefit Analysis

The cost-benefit analysis illustrates the financial and strategic value of the DOST STARBOOKS: Whiz Challenge Project. This helps decision-makers assess whether the project is worth pursuing. Since this is a student-led initiative using existing resources and open-source tools, the direct costs are minimal, while the benefits are significant.

Action	Action Type	Description	First year costs (- indicates anticipated savings)
Platform development	Cost	Student-led creation of the quiz system	₱0
Tools and software	Cost	Open-source tools for web development and data tracking	₱0
Deployment	Cost	Use of existing STARBOOKS kiosks and library computers	₱0
Increased student engagement	Benefit	Gamified quizzes encourage participation, equivalent to higher library usage and learning efficiency. Estimated value: ₱200,000	-₱200,000
Improved learning outcomes	Benefit	Students receive instant feedback; reduces tutoring/review sessions cost. Estimated value: ₱150,000	-₱150,000
Data-driven decision-making	Benefit	Administrators track performance efficiently, reducing manual tracking workload. Estimated value: ₱50,000	-₱50,000
Expanded outreach	Benefit	Ability to reach more students efficiently through existing platform. Estimated value: ₱100,000	-₱100,000
Net First Year Savings			**₱500,000**

The DOST STARBOOKS: Whiz Challenge provides high impact at zero financial cost, with estimated benefits of ₱500,000 in the first year, making it a highly valuable and cost-effective educational initiative.

7. Alternatives Analysis

While the selected option is the student-led DOST STARBOOKS: Whiz Challenge platform, other alternatives were considered. The reasons for not selecting them are summarized below.

No Project (Status Quo)	**Reasons For Not Selecting Alternative**
Keep the mainframe legacy system in place	• Student engagement remains low • No improvements in learning outcomes • Manual tracking of quiz performance continues • Limited outreach and data analysis
Alternative Option	**Reasons For Not Selecting Alternative**
Minor Updates Only	• Small improvements do not address core issues like gamified participation or real-time feedback • Existing system limitations remain
Fully Online System	• Not all students have stable internet connectivity • Access equity issues; may exclude students without personal devices
Third-party Platform	• High cost of subscription or licensing • Limited customization to fit STARBOOKS library activities and branding • Less student ownership and involvement in development

8. Approvals

The signatures of the people below indicate an understanding in the purpose and content of this document by those signing it. By signing this document you indicate that you approve of the proposed project outlined in this business case and that the next steps may be taken to create a formal project in accordance with the details outlined herein.

Approver Name	Title	Signature	Date
Narquita, M.	ISS Head, DOST-STII		04/08/2026
Dumbrique, K.	Project Manager		04/07/2026